DILSON BHARATH R

AI Enthusiast & Software Developer

Email: dilsonbharath76@gmail.com

LinkedIn: LinkedIn

GitHub: GitHub

LeetCode: LeetCode

Website: Website

SUMMARY

Final-year Artificial Intelligence and Data Science student with strong programming, machine learning, web development skills and continuously learning and exploring the **evolving AI technologies**. Seeking opportunities to apply **Python** and **AI** expertise to build data-driven systems, with a focus on impactful technology solutions.

INTERNSHIPS

Innovative Engineer Feb 2025 - Jun 2025
Forge Innovation & Venture • Coimbatore

During my internship, I **led a team** of four in developing an **OpenCV-based pallet detection** system to identify position and alignment accurately. I was responsible for coordinating tasks, managing team communication, and reporting progress to senior professionals. I contributed to **dataset collection, preprocessing, and feature vector preparation** for deep learning integration. The system leveraged computer vision techniques to improve detection reliability, and our project was recognized for strong execution. Additionally, I gained practical **exposure to AI tools**, designing and professional IT workflows.

EDUCATION

B.Tech in Artificial Intelligence & Data Science Nov 2022 - Present
Kumaraguru College of Technology • Coimbatore

CGPA: 7.03

HSC Sep 2021 - May 2022
SMA Matriculation Higher Secondary School

Percentage: 95.33%

SSLC Jun 2019 - Mar 2020
SMA Matriculation Higher Secondary School

Percentage: 96.4%

SKILLS

Programming Languages : Python, JavaScript (Intermediate), C++ (Basics)

Database : MySQL

Web Development : HTML, CSS, Bootstrap, React, FastAPI

AI : Machine Learning, RAG

Soft Skills : Team Collaboration, Communication, Adaptability, Problem Solving

PROJECTS

Habit Tracker Application with ChatBot
React, FastAPI, MySQL, RAG

This project is a **full-stack** personal productivity platform that unifies habit tracking, journaling, expense management, skill challenges, and progress analytics in a single experience. In this project I used techstack for frontend React framework and backend I used FastAPI and then database used MySQL. Its standout feature is a **RAG chatbot** that combines knowledge-base retrieval search and Hugging Face LLM(mistralai/Mistral-7B-Instruct-v0.3) I choose this because this model carries **light weight with best potential and fast** inference to generate contextual, personalized responses.

ML-Based Insider Threat Detection System
CatBoost Model, Hyperparameter Tuning

Developed a machine learning–based insider threat detection module for a **healthcare EHR system** by analyzing user activity logs of doctors, nurses, and staff to identify anomalous behavior. Collected and engineered relevant behavioral features from access patterns, timestamps, and record interactions to build a robust dataset and support future scalability of the model. Built and optimized a **CatBoost classification** model, improving performance from **91% to 95%** accuracy through systematic **hyperparameter tuning**. Designed an end-to-end ML pipeline including data preprocessing, feature engineering, model training, validation, and evaluation. Integrated the trained model with the backend to enable efficient anomaly detection within the application workflow. **Collaborated with the full-stack development team** to align machine learning outputs with frontend, backend, and database components for seamless system integration.

Portfolio Website
CSS, React

Developed a personal portfolio website using **ReactJS and CSS to showcase projects**, technical skills, and achievements in a structured and responsive interface. Built the project as a self-initiated effort to establish a **professional online presence** and demonstrate frontend development capabilities.

CERTIFICATIONS

Machine Learning with Python
IBM • Nov 2025

Generative AI Fine-Tuning for LLMs
IBM • Nov 2025

Fundamentals of AI Agents Using RAG and LangChain
IBM • Nov 2025

AWARDS & ACHIEVEMENTS

- **100+ Days Batch** - award for problem solving on Leetcode.
- **Anna University Zone XI Champions 2024** - leading the team front and made the important points at the crucial stages for winning.

INTERESTS

Problem Solving • Chess • Sports • Travel